

Khoi (Ben) Nguyen

Villanova, PA | benkhoi2006@gmail.com | (667) 310 9940 | <https://www.linkedin.com/in/khoi-ben-nguyen/>

EDUCATION

Villanova University - College of Engineering, Villanova, PA Aug 2024 – May 2028

Bachelor of Science in Mechanical Engineering, Mechatronics Minor | **GPA: 4.00/4.00**

- **Coursework:** Thermodynamics, Dynamics System, Mechanics of Materials, Electric Circuits, Calculus III, Differential Equations, Computer Program for Mech Eng, Communication, Macroeconomics, Stats Method
- **Awards/Honors:** Dean's List (*All terms*), Undergraduate Research Scholars
- **Involvement (Villanova):** Badminton Club **President**, Formula SAE Driver Control System, Special Olympics Local Program Host, Villanova American Society of Mechanical Engineers

EXPERIENCE

Formula SAE – Driver Control System (Pedal Box) | Villanova University Jan 2026 – Present

- Designing Villanova's FSAE (VU18) car's **pedal box** in **SolidWorks**, focusing on packaging & adjustability
- **Manufactured** components for current car: including pedal box parts, steering wheel components, and 3D-printed molds for composite layups using **milling**, **lathing**, and additive manufacturing
- Designed a pedal-box **return spring system** in **SolidWorks**; used a **MATLAB** script to calculate minimum force/deflection for **20°** pedal travel; Built a **BOM** by selecting off-the-shelf components to meet constraints

3D Lab: Revolutionizing Research with 3D Printing | Villanova University Aug 2025 - Present

ME Sophomore Research Scholars - Undergraduate Researcher

- **Designed** and **built** a benchtop **tensile-testing stage** with **8 custom components**; integrating a **TMC2209** stepper driver, **ADS1115**, and **load cell** system; wrote **Python control software** to **automate** testing and export real-time data (time, displacement, load ± 0.03 N, and step count) to **CSV** for analysis
- Developing a repeatable hydrogen-embrittlement testing workflow for steel to enable repeatable tensile testing and reliable material behavior and crack characterization

Engineering Lab Assistant – Drosdick Innovation Lab | Villanova University Aug 2025 - Present

Lab Assistant

- Support, manage, and provide guidance on **3D printing**, **soldering**, **woodworking**, makerspace tools to students and faculties
- Maintain and modify **20** Bambu printers, **2** resin printers, **1** laser cutter, woodworking shop, power tools

INNOVATE: L3Harris Summer Program | Villanova University June 2025 - July 2025

Mechanical Engineering Lead

- Led a team of four to develop **AeroVision**, a low-cost CFD tool using **Machine Learning** and **Computer Vision** to visualize airflow over airfoils
- **3D modeled and built** a motorized wind tunnel positioning system in **SolidWorks** and conducted **100+ flow visualization tests** in Villanova's **closed-return wind tunnel** to generate training data

PROJECTS

Voron 2.4 and 0.2 3D Printers Construction

- Independently built custom **Voron printers** from scratch, enhanced, and calibrated **hardware and software**
- Gained extensive experience in troubleshooting **extrusion**, **adhesion**, and **firmware tuning**

DIY CNC Plasma Cutter

- Built a DIY CNC plasma cutter using 3D-printed and off-the-shelf components for sheet metal fabrication
- Completed **wiring**, **troubleshooting**, and **Fusion 360 CAM tuning** to enhance Formula SAE production of tabs, brackets, and other metal parts

SKILLS

Software & CAD: SolidWorks, Fusion 360, Microsoft Office, Python, MATLAB, OrcaSlicer

Lab Tools: 3D printers, Resin printing, Milling & Lathing machine, Laser cutter, Instron 5980 Tensile tester, Closed-return/Open wind tunnel, Markforged Mark Two, Woodworking machinery